



Project Final Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	25
Problem understanding & Analysis	7					
Implementation	8					
Report writing	10					
Total obtained mark						
Comments						

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Course Teacher Name: Mr. Humaun Kabir Nayem

Designation: Lecturer

Name	ID	Section
Farhan Ibn Mustafa	0242310005341325	A2
Yeasmin Akter	0242310005341207	A2
Md Mobin	0242310005341226	A2

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Psychological Support System (PSS)

Introduction

Mental health is a key aspect of our overall well-being, yet many individuals struggle to locate the correct help when they need it the most. The Psychological Support System (PSS) is a conceptual initiative that aims to offer a secure, accessible, and organized environment where people can get mental health treatment. This digital platform intends to provide AI-powered initial evaluations, professional counseling directories, peer support networks, and self-help tools to assist users in navigating their mental health journey. This project is being created as part of a System Analysis and Design Capstone Project, emphasizing system documentation, workflow design, and functional visualization rather than real implementation or coding. Our mission is to knock down challenges to mental health concerns and build a system that fulfills the needs of individuals, whether they are students, employees, or anybody dealing with emotional difficulties.

Problem Statement

Plenty of people desire to get assistance with their mental health, yet they face difficulties like:

- **Limited Accessibility:** Finding a professional therapist or counselor can be difficult and expensive.
- **Stain & Privacy Concerns:** Many fear being judged for seeking mental health support.
- **Lack of Peer Support:** The absence of a strong support network can make struggles feel isolating.
- **Resource Constraints:** There is a shortage of mental health professionals, leading to long wait times for therapy sessions.

The PSS seeks to close these gaps by offering an automated, centralized support system that includes AI-powered assistance, professional counseling links, peer groups, and self-help materials.

Objectives

The Psychological Support System seeks to:

- Provide chatbot assistance driven by AI to offer preliminary mental health evaluations and counseling.
- To provide expert assistance, compile a list of certified counselors.
- Create a forum for peer support where people can interact and exchange stories.
- Offer self-help resources like journaling activities and meditation manuals.
- Include resources for emergencies, such as hotlines and ways to get help right away.

Scope

The PSS will prioritize the following major areas:

- **AI Chatbot Support:** A virtual assistant that helps users identify their mental health needs.
- **Professional Counselor Directory:** A structured list of certified professionals, with appointment booking options.
- **Peer Support Forum:** A safe space for discussions and emotional support from others who understand.
- **Self-Help Resources:** A collection of mindfulness exercises, therapy guides, and educational content.
- **Emergency Contact Integration:** Direct access to crisis helplines for urgent support.

Key Features

1. User Registration and Login:

- Patients and support professionals can securely register and log in.
- Password recovery options are available for user convenience.

2. Profile Management:

- Patients can update their profiles with personal details and preferences.

- Support professionals manage their expertise and availability profiles.

3. Self-Assessment:

- Patients can select their mood and answer a set of questions for self-assessment.
- Patients can track their progress and build streaks by completing self-assessments daily, promoting mental health consistency.

4. Appointment Scheduling:

- Patients can check available slots and book appointments with support professionals.
- Professionals can confirm or reject appointment requests.

5. Session Management:

- Patients and support professionals are notified once the appointment is confirmed.
- The system helps in managing session timings and flow.

6. Diagnosis and Support:

- Support professionals assess the patient's mental health, provide guidance, recommend exercises, or refer the patient to other specialists if necessary.

7. Group Discussions & Forum Participation:

- Patients can participate in open forum discussions where they can engage with others.
- Group discussions allow patients to interact in a supportive group environment.
- Patients can also join anonymous discussions if they prefer to remain anonymous while seeking support.

8. Chatbot for Quick Suggestions:

- A chatbot is available to provide immediate suggestions or answers for patients seeking quick assistance or guidance.

9. Feedback System:

- Patients can provide feedback after each session to help professionals improve their services.

10. Billing and Payment:

- After the session, a bill is generated, and patients are notified of the payment.
- Multiple payment options are available for patients to complete their payments.

11. Helpline Support:

- 24/7 helpline support is available for urgent issues, with a comprehensive FAQ section for quick answers.

Scenario Writing

Scenario writing involves describing specific situations or use cases that illustrate how a system or software application will be used by its users. These scenarios help to better understand the system's functionalities, interactions, and outcomes. In the context of the Psychological Support System, let's write some scenarios to depict typical user interactions:

Scenario-1: Register Account

Scenario Description:

- Patient or support professional registers by providing necessary details.
- The system validates the registration and provides access to the dashboard.

Scenario-2: Complete Daily Self-Assessment

Scenario Description:

- The patient logs in and selects their mood from a list.
- The patient answers a series of questions related to their emotional state.
- The system records the answers and adds to the streak if done daily.

Scenario-3: Book an Appointment

Scenario Description:

- The patient logs in and searches for available support professionals.
- The patient selects an available time slot and submits an appointment request.
- The support professional either accepts or rejects the request, and the patient is notified.

Scenario-4: Chat with Bot for Quick Suggestion

Scenario Description:

- The patient opens the chatbot feature.
- The patient asks a question or seeks quick suggestions regarding their mental health.
- The bot provides an answer or suggestion based on the patient's inquiry.

Scenario-5: Attend Consultation

Scenario Description:

- The patient joins the scheduled session.
- The support professional conducts the session, providing guidance or assistance.
- The session ends with a request for patient feedback.

Scenario-6: Submit Feedback

Scenario Description:

- After the session, the patient is asked to provide feedback on the consultation.
- The feedback is submitted and stored for the professional to review.

Scenario-7: Make Payment

Scenario Description:

- The patient receives a payment notification after the session.
- The patient selects a payment method and completes the payment.
- A receipt is generated, and the patient is notified of successful payment.

Scenario-8: Join Group Discussion

Scenario Description:

- The patient joins a group discussion related to mental health and wellness.
- The patient can share personal experiences, ask for advice, or support others.

Scenario-9: Participate in Anonymous Discussion

Scenario Description:

- The patient chooses to remain anonymous and participates in a discussion without revealing their identity.
- The system ensures that no personal information is shared during the anonymous participation.

Stakeholders

The key stakeholders involved in this Hotel Management System are:

1. Patient:

- Patients use the system to book appointments, attend sessions, participate in self-assessments, chat with the bot for quick suggestions, and track their progress.
- They can join discussions in forums, group chats, or anonymously.
- They provide feedback after sessions and make payments for services.

2. Support Professional (Psychologist, Psychiatrist):

- Support professionals manage their availability, accept or reject appointments, conduct consultations, and provide guidance.
- They review patient feedback to improve their services.

3. Admin (System Administrator):

- The admin manages system operations, user access, security measures, and ensures smooth functionality.
- They handle any user support queries and maintain system stability.

4. Payment Gateway Provider:

- Third-party services handle payment processing for patients.

Feasibility Study

The Psychological Support System (PSS) proposes a comprehensive solution to address the challenges and demands of modern mental health support. This feasibility study aims to assess the viability of implementing the PSS by analyzing its technical, operational, economic, and scheduling aspects.

1. Technical Feasibility:

a. Hardware Compatibility:

The PSS should be compatible with commonly used devices such as smartphones, tablets, and

personal computers. A review of user accessibility and potential hardware limitations will be conducted to ensure widespread usability.

b. Software Compatibility:

Ensure that the system integrates seamlessly with existing software environments and platforms. Compatibility with web browsers, mobile operating systems (iOS, Android), databases, and third-party services such as chatbot APIs and video conferencing tools will be evaluated.

c. Technical Expertise:

Assess the availability of skilled personnel or the need for training to operate and maintain the PSS. Required expertise in web development, data management, and mental health compliance will be identified and addressed

2. Operational Feasibility:

a. User Acceptance:

Gather feedback from potential users, including patients, mental health professionals, and support moderators, to ensure the system aligns with their needs and expectations. Features such as anonymity, chatbot interaction, self-assessment, and resource access are expected to enhance user acceptance.

b. Impact on Current Operations:

Analyze how the implementation of the PSS will affect existing counseling and support workflows. Mitigation strategies for any disruption in service delivery or professional workflows will be identified, including gradual rollout and training.

3. Economic Feasibility:

a. Cost-Benefit Analysis:

Conduct a thorough analysis of the costs associated with developing, implementing, and maintaining the PSS against the expected benefits, such as increased mental health access, reduced manual workloads, improved patient satisfaction, and potential for scaled institutional use.

b. Return on Investment (ROI):

Calculate the projected ROI over a specified period, considering both tangible benefits (e.g., reduced operational costs, scalability) and intangible benefits (e.g., better user engagement, societal mental health improvement).

4. Scheduling Feasibility:

a. Project Timeline:

Develop a realistic timeline for the different phases of the PSS implementation, including system planning, development, testing, deployment, and training. Consideration will be given to testing phases and feedback integration.

b. Dependencies:

Identify dependencies on external factors, such as third-party service integration (e.g., chatbots, mood analysis APIs) or regulatory approvals for data protection and mental health guidelines, that may impact the project schedule.

The feasibility study concludes that the Psychological Support System is technically, operationally, economically, and schedulably viable. The benefits of enhanced mental health accessibility, user engagement, and potential for broad implementation outweigh the costs associated with development and deployment. Recommendations for mitigating risks and maximizing benefits will be incorporated into the project plan to ensure successful PSS deployment.

Software Requirements Specification

FR01	Registration
Description	Users (patients, psychologists, psychiatrists) must register before accessing the system.
Stakeholder	Patient, Psychologist, Psychiatrist

FR02	Login
Description	Registered users must log in to access features like assessment, sessions, and chat.
Stakeholder	Patient, Psychologist, Psychiatrist, Admin

FR03	Add Profile Info
Description	Users complete their profiles with relevant personal and professional information.
Stakeholder	Patient, Psychologist, Psychiatrist

FR04	Self-Assessment
Description	Patients perform mood-based assessments to receive personalized suggestion
Stakeholder	Patient

FR05	Daily Mood Logging
Description	Patients can log daily mood data to build a streak.
Stakeholder	Patient

FR06	Chatbot Interaction
Description	Patients can interact with a bot to receive quick support or coping strategies.
Stakeholder	Patient

FR07	Book Counseling Session
Description	Patients can book appointments with available psychologists or psychiatrists
Stakeholder	Patient

FR08	Session Confirmation
Description	Psychologists/Psychiatrists confirm or reschedule booked appointments.
Stakeholder	Psychologist, Psychiatrist

FR09	Conduct Session
Description	Therapist conducts the session and provides notes, suggestions, or prescriptions.
Stakeholder	Psychologist, Psychiatrist

FR10	View Suggestions
Description	Patients can view AI-generated or expert-given suggestions.
Stakeholder	Patient

FR11	Forum Discussion
Description	Patients can participate in public or anonymous discussion forums.
Stakeholder	Patient

FR12	Group Chat
Description	Group chat sessions allow shared experience discussions
Stakeholder	Patient

FR13	Anonymous Chat
Description	Patients may post and discuss anonymously to maintain privacy.
Stakeholder	Patient

FR14	Add Feedback
Description	Patients provide session or system feedback.
Stakeholder	Patient

FR15	Manage Feedback
Description	Admin monitors feedback and flags inappropriate content
Stakeholder	Admin

FR16	Payment Processing
Description	Patients pay for booked sessions securely.
Stakeholder	Patient

FR17	Emergency Contact
Description	Patients can access emergency contacts or request urgent help.
Stakeholder	Patient

FR18	Notification System
Description	Notifications for appointments, session starts,

	and bot alerts.
Stakeholder	Patient, Psychologist, Psychiatrist

FR19	Report Generation
Description	Admin can generate reports on usage, sessions, and mood trends.
Stakeholder	Admin

FR20	Manage Users
Description	Admins can manage registered users and suspend suspicious activity.
Stakeholder	Admin

User Case Descriptions

Case Description-01: Self-Assessment

Use Case	Self-Assessment
Goal	Patient completes a mood-based assessment to understand their mental state and get

	suggestions.
Precondition	Patient must be logged into the PSS system.
Success End Condition	Notification: "Assessment Completed Successfully!"
Primary Actors Secondary Actors	Patient None
Trigger	Patient selects "Self-Assessment" from the dashboard.
Description / Main Success Scenario	1. Patient visits the dashboard. 2. Clicks "Self-Assessment. 3. Selects mood from a list (e.g., Happy, Sad). 4. Answers questions about emotional state. 5. Clicks "Submit." 6. System saves responses and displays suggestions. 7. Notification: "Assessment Completed Successfully!"
Alternative Flows	2.1. System error. 2.1.a. Try again later. 4.1. Patient skips questions. 4.1.a. System prompts: "Please answer all questions." 5.1. Submission fails.

	5.1.a. Notification: "Try again."
Quality Requirements	Assessment should be completed within 5-10 minutes.

Case Description-02: Book Counseling

Use Case	Book Counseling
Goal	Patient books an appointment with a psychologist or psychiatrist.
Precondition	Patient must be logged in and have access to the counselor directory.
Success End Condition	Notification: "Appointment Booked Successfully!"
Failed End Condition	Notification: "Booking Failed."
Primary Actors	Patient
Secondary Actors	Psychologist, Psychiatrist
Trigger	Patient selects "Book Counseling" from the dashboard.

Description / Main Success Scenario	1. Patient visits the dashboard. 2. Clicks "Book Counseling." 3. Browses available counselors. 4. Selects a counselor and time slot. 5. Submits booking request. 6. System notifies counselor. 7. Counselor confirms. 8. Notification: "Appointment Booked Successfully!"
Alternative Flows	3.1. No counselors available. 3.1.a. System suggests alternative slots. 5.1. System error. 5.1.a. Try again later. 7.1. Counselor rejects. 7.1.a. Patient is notified to choose another slot.
Quality Requirements	Booking should take 3-5 minutes.

Case Description-03: Chatbot Interaction

Use Case	Chatbot Interaction
Goal	Patient interacts with AI chatbot to get instant mental health suggestions.

Precondition	Patient must be logged into the PSS system.
Success End Condition	Notification: "Chatbot Response Delivered!"
Failed End Condition	Notification: "Chatbot Unavailable."
Primary Actors	Patient
Secondary Actors	None
Trigger	Patient opens the chatbot feature from the dashboard.
Description / Main Success Scenario	1. Patient visits the dashboard. 2. Clicks "Chatbot." 3. Types a question or describes an issue. 4. Clicks "Send." 5. Chatbot processes and responds with suggestions. 6. Notification: "Chatbot Response Delivered!"
Alternative Flows	3.1. Query unclear. 3.1.a. Chatbot prompts: "Please clarify your question." 5.1. System error. 5.1.a. Notification: "Try again later."

Quality Requirements	Chatbot should respond within 15-30 seconds
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Case Description-04: Daily Mood Logging

Use Case	Daily Mood Logging
Goal	Patient logs daily mood to track emotional progress.
Precondition	Patient must be logged into the PSS system.
Success End Condition	Notification: "Mood Logged Successfully!"
Failed End Condition	Notification: "Mood Not Logged."
Primary Actors	Patient
Secondary Actors	None
Trigger	Patient selects "Daily Mood Logging" from the dashboard.
Description / Main Success Scenario	1. Patient visits the dashboard. 2. Clicks "Daily Mood Logging." 3. Selects mood from a list (e.g., Anxious, Calm). 4. Optionally adds notes. 5. Clicks "Submit."

	6. System saves log and updates streak. 7. Notification: "Mood Logged Successfully!"
Alternative Flows	3.1. No mood selected. 3.1.a. System prompts: "Please select a mood." 5.1. Submission fails. 5.1.a. Notification: "Try again."
Quality Requirements	Mood logging should take 1-2 minutes.

Case Description-05: Forum Discussion

Use Case	Forum Discussion
Goal	Patient participates in public or anonymous discussions for support.
Precondition	Patient must be logged into the PSS system.
Success End Condition	Notification: "Post/Comment Submitted Successfully!"
Failed End Condition	Notification: "Submission Failed."
Primary Actors	Patient
Secondary Actors	None
Trigger	Patient selects "Forum Discussion" from the dashboard

Description / Main Success Scenario	1. Patient visits the dashboard. 2. Clicks "Forum Discussion." 3. Chooses public or anonymous mode. 4. Posts a message or comments on a thread. 5. Clicks "Submit." 6. System saves and displays post in forum. 7. Notification: "Post/Comment Submitted Successfully!"
Alternative Flows	4.1. Empty post. 4.1.a. System prompts: "Please enter a message." 5.1. System error. 5.1.a. Notification: "Try again later."
Quality Requirements	Posting should take 2-3 minutes.

Case Description-06: Add Feedback

Use Case	Add Feedback
Goal	Patient provides feedback after a session or system use.
Precondition	Patient must have completed a session or used a feature.
Success End Condition	Notification: "Feedback Submitted"

	Successfully!"
Failed End Condition	Notification: "Feedback Not Submitted."
Primary Actors	Patient
Secondary Actors	Psychologist, Psychiatrist, Admin
Trigger	System prompts "Add Feedback" after a session.
Description / Main Success Scenario	1. Patient completes a session. 2. System prompts "Add Feedback." 3. Patient rates session (e.g., 1-5 stars). 4. Optionally adds comments. 5. Clicks "Submit." 6. System saves feedback for review. 7. Notification: "Feedback Submitted Successfully!"
Alternative Flows	3.1. No rating provided. 3.1.a. System prompts: "Please provide a rating." 5.1. Submission fails. 5.1.a. Notification: "Try again."
Quality Requirements	Feedback submission should take 1-2 minutes.

User Profile Descriptions

User Profile-01: Patient

User Class	Notes on Characteristic	Requirement Implied
Type of User	Patient	Verification
Age Range	18-65	Verification
Frequency of Use	Daily or as needed (e.g., for mood logging, sessions)	Performance, Operation, Acceptance
Mandatory	No	
Computer Experience	Basic to moderate (can use apps, websites)	Usability, Documentation
Education	High school or above	
Goals	Access mental health support, track progress, join forums	Resource, Performance, Security, Acceptance, Usability
Language Skills	English, local languages (e.g., Bangla	Localization
Number of Users	100-1000	Scalability, Performance, Operation

Training	Minimal, prefers an intuitive interface	Documentation, Usability
Other Systems Used	Social media, health apps	Compatibility
Ways of Working	Self-paced, seeks privacy and ease	Acceptance, Security, Usability, Maintenance

Project Block Diagram

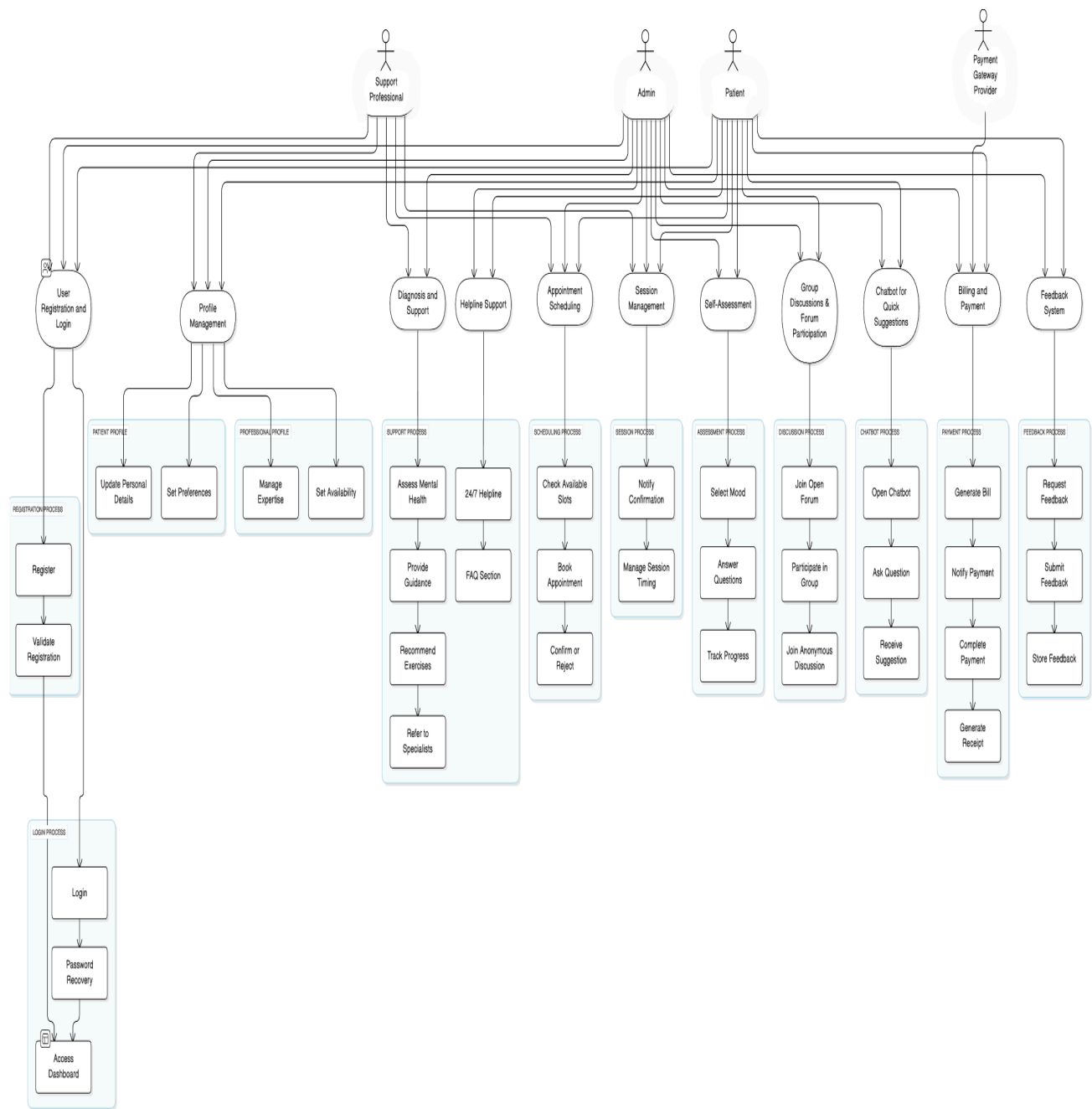


Figure-1: Block Diagram for Psychological Support System

User Case Diagram

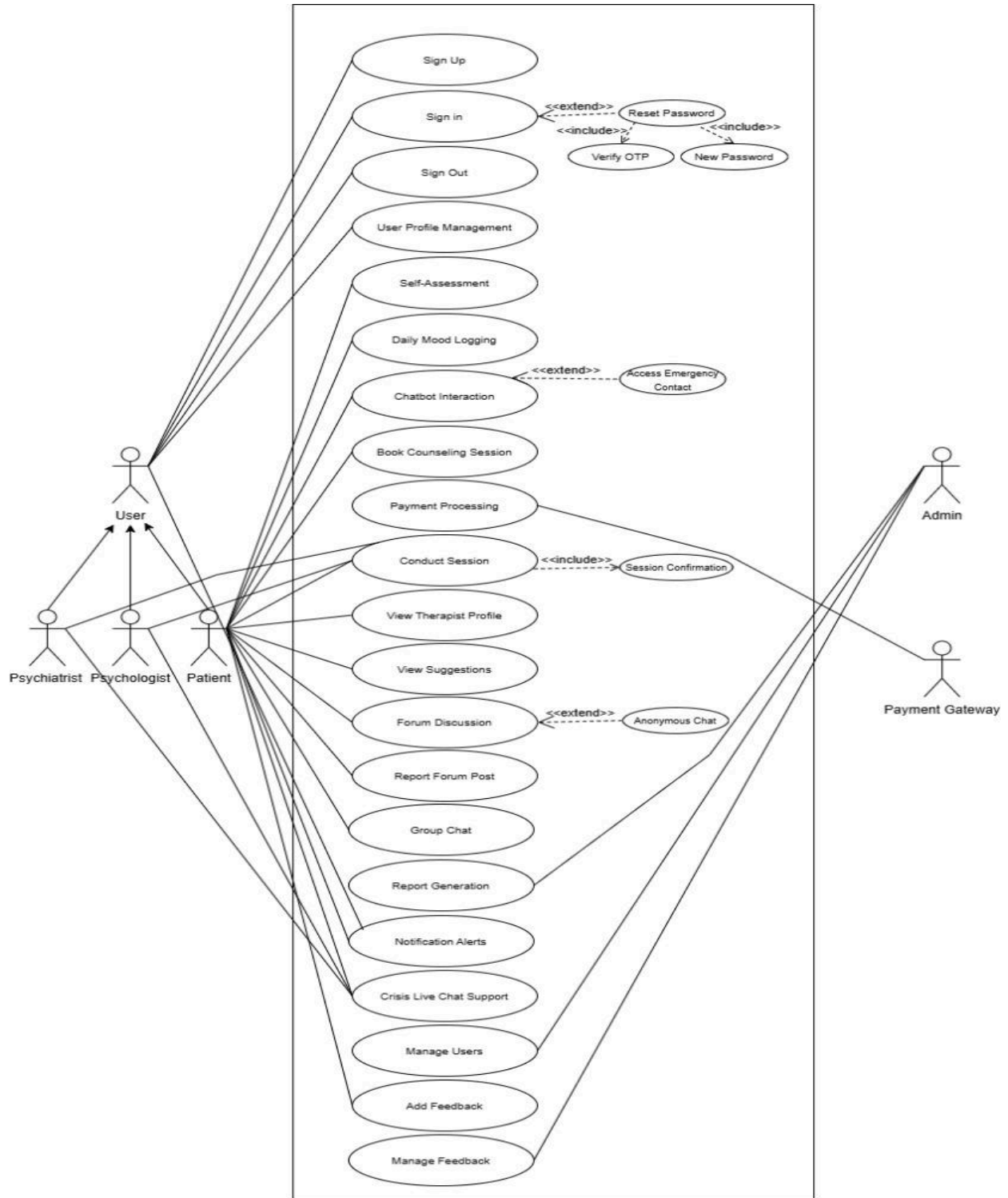


Figure-2: Use Case Diagram for Psychological Management System

System Design

Data Flow Diagram

Data Flow Diagram Level-0:

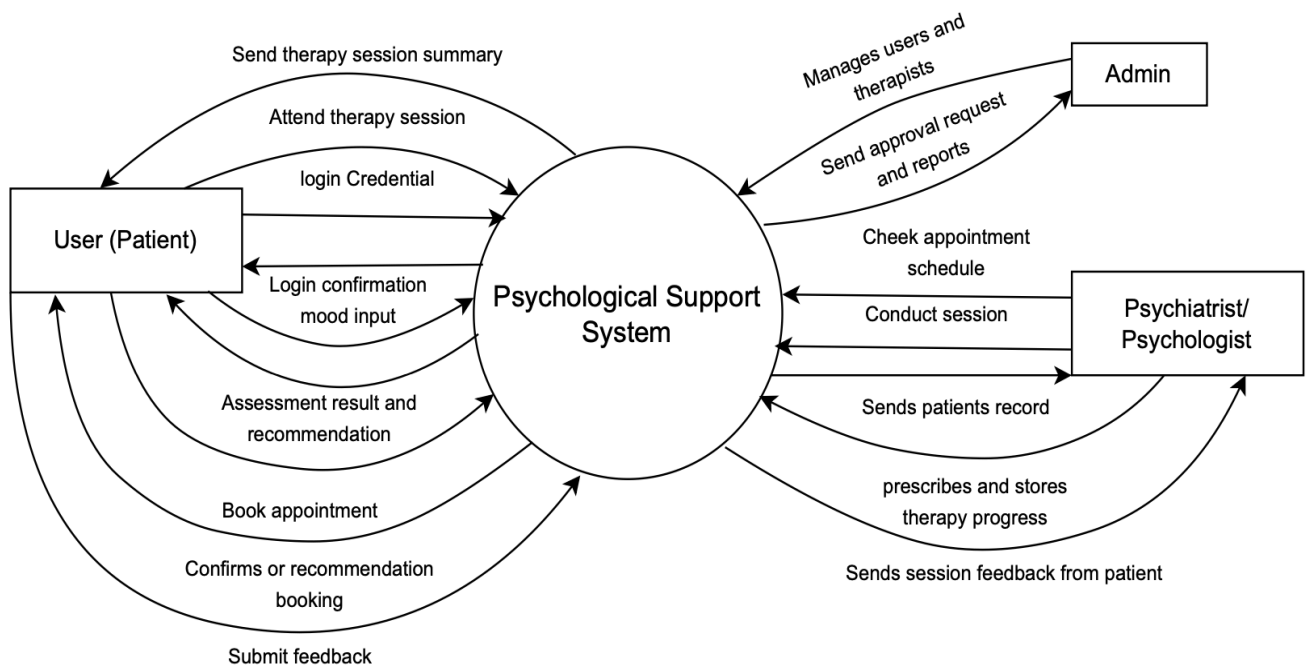


Figure-3: Data Flow Diagram Level 0

Data Flow Diagram Level-1:

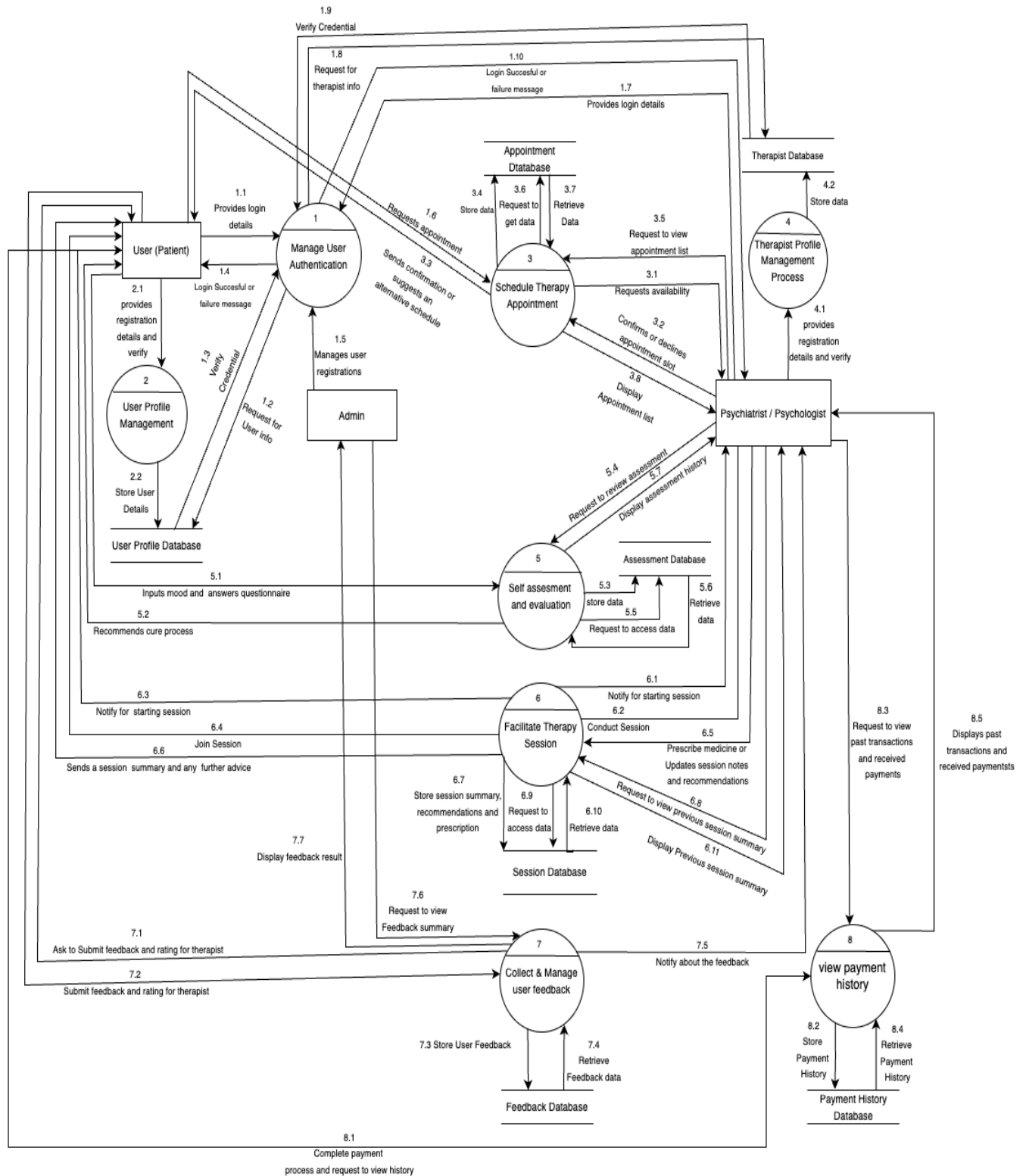


Figure-4: Data Flow Diagram Level 1

Activity Diagram

Activity Diagram-1: Psychiatrist

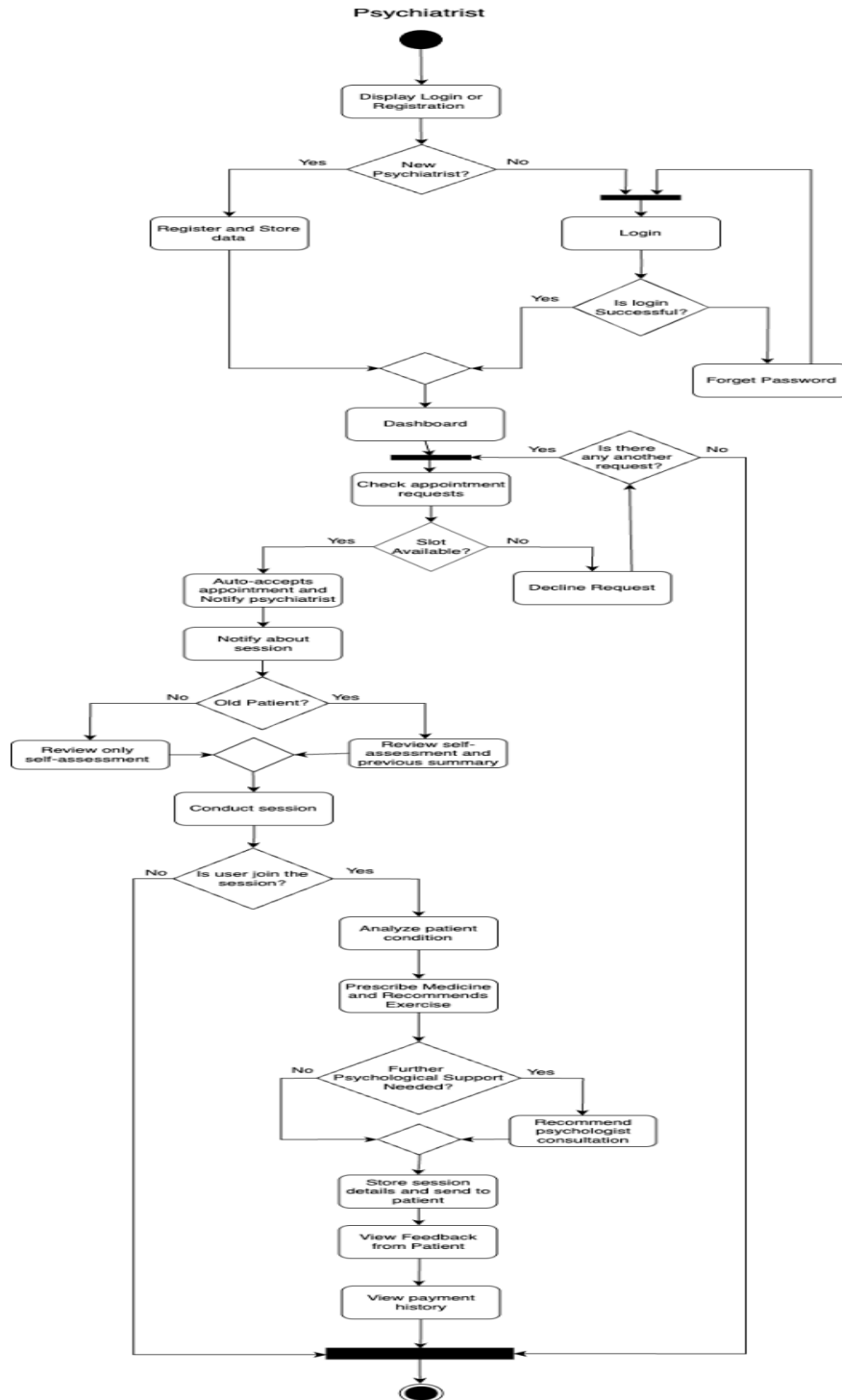


Figure-5: Activity Diagram for Psychiatrist

Activity Diagram-2: Psychologist

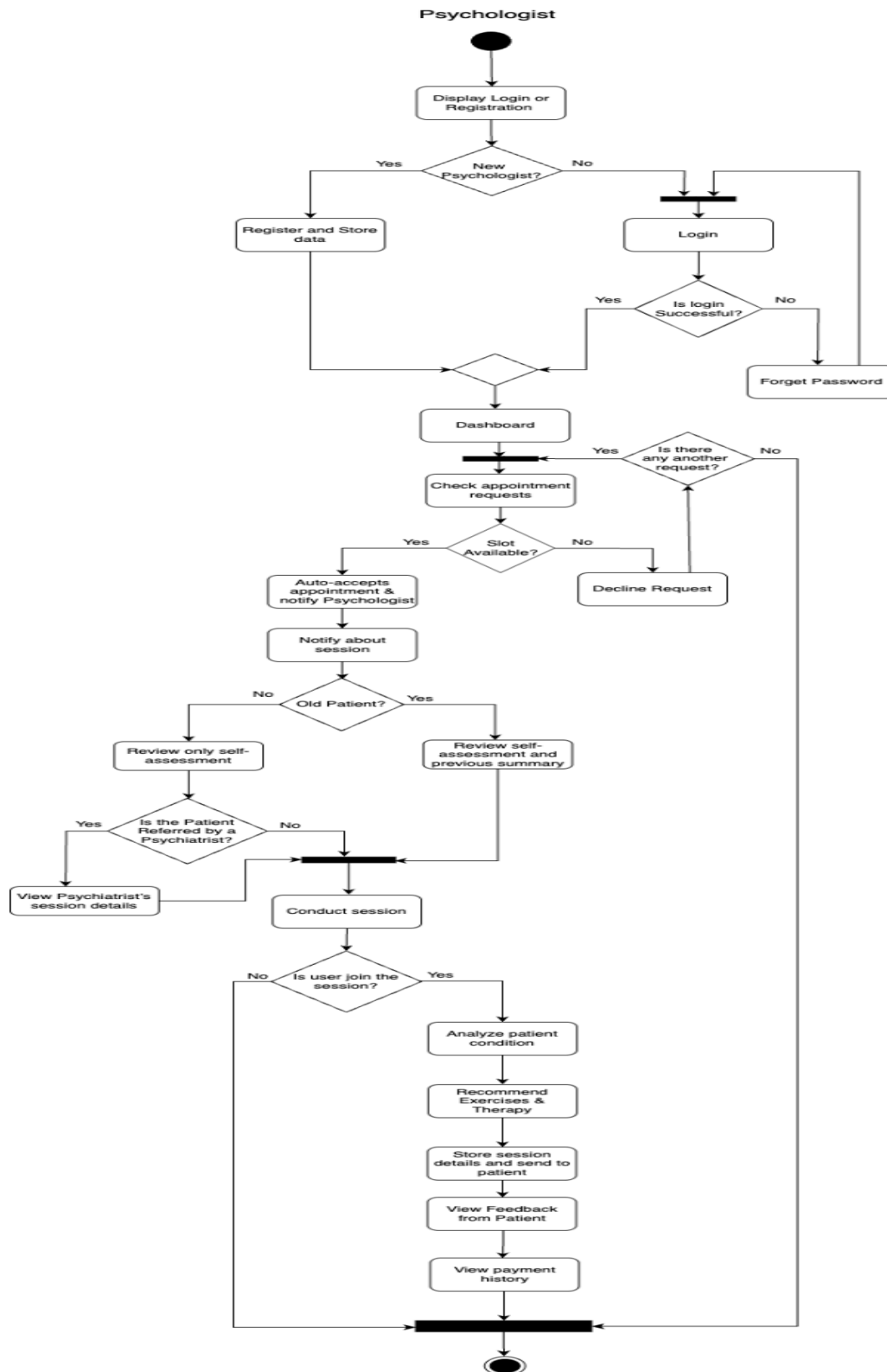


Figure-6: Activity Diagram for Psychologist

Activity Diagram-3: Patient

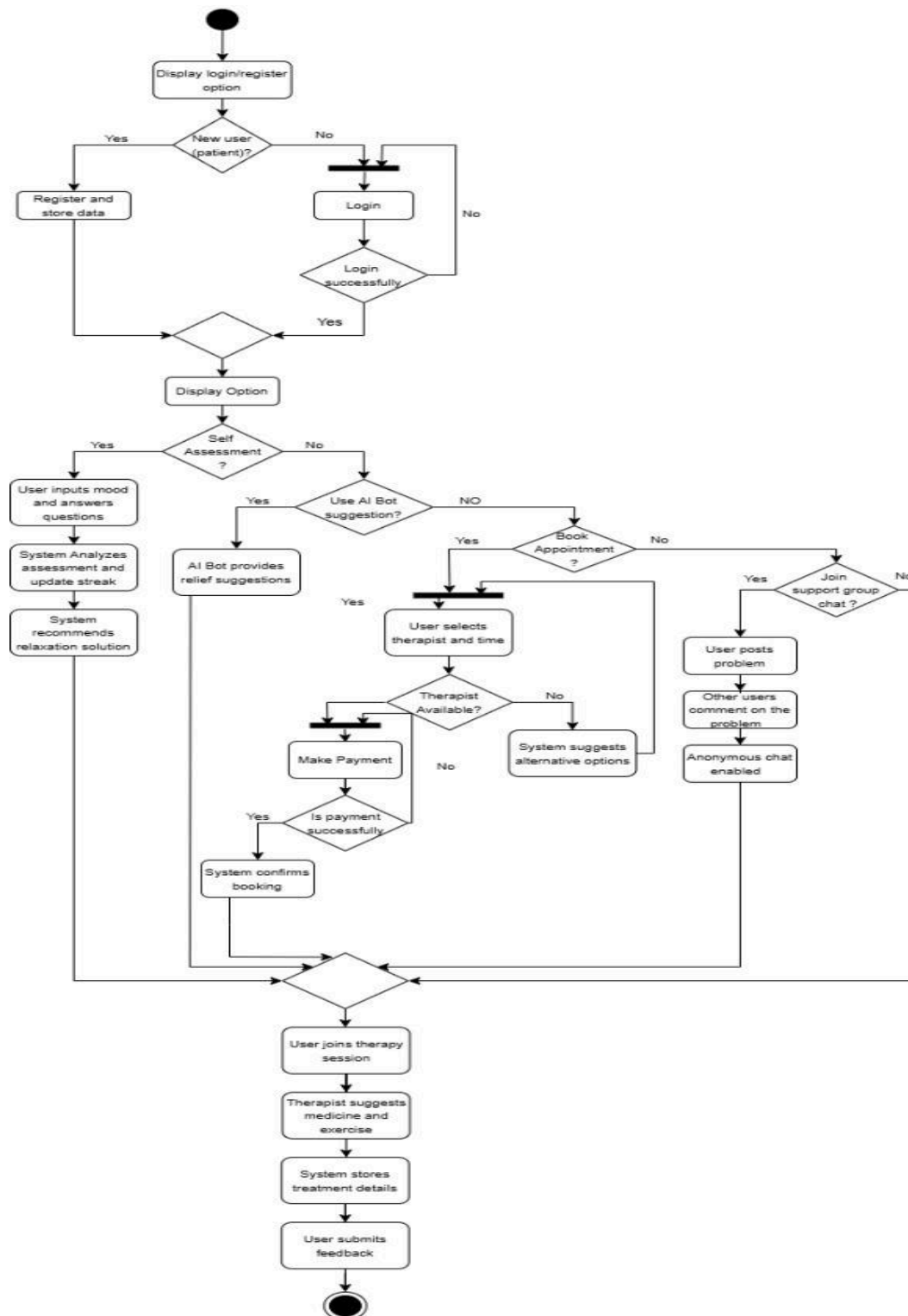


Figure-7: Activity Diagram for Patient

Activity Diagram-4: Admin

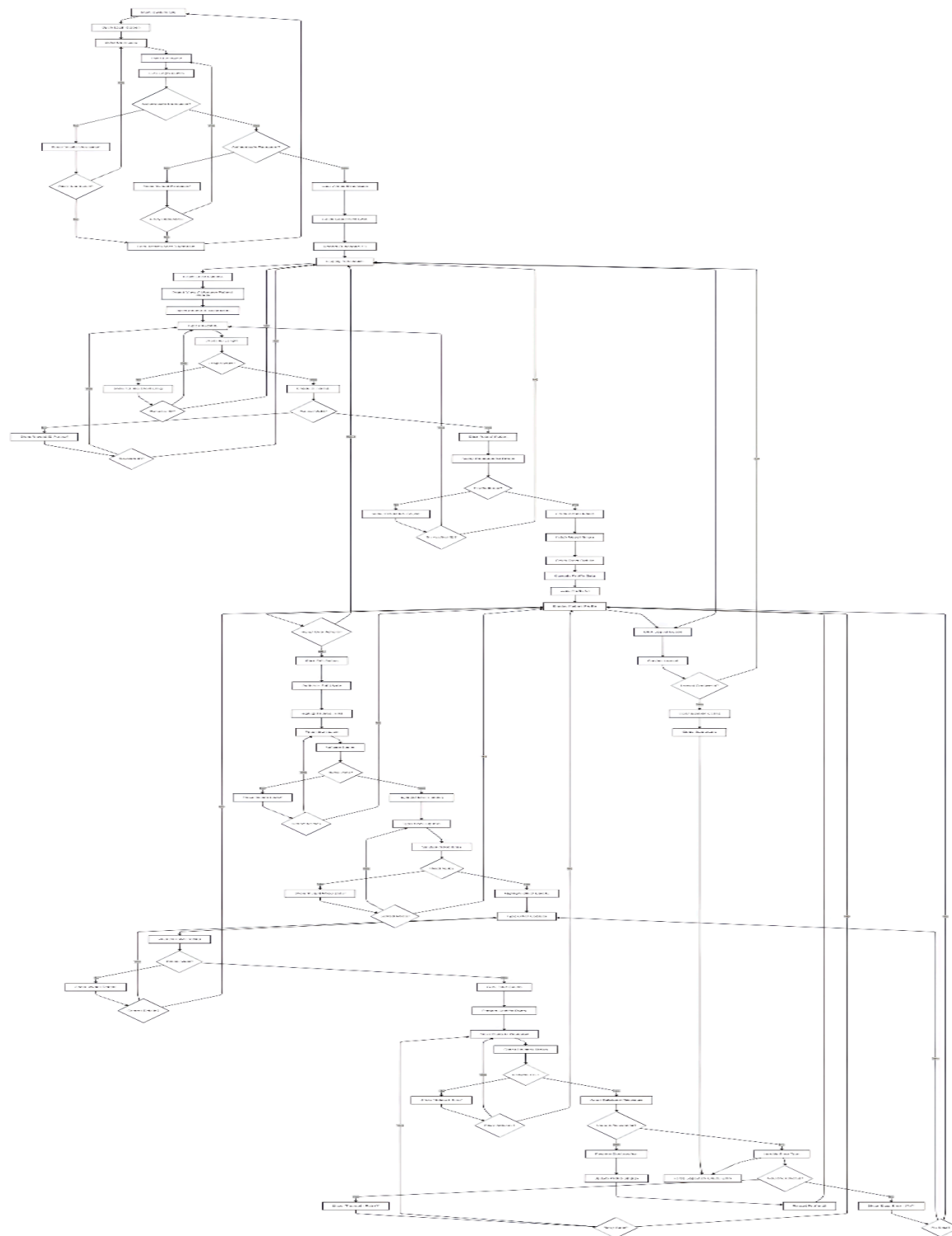


Figure-8: Activity Diagram for Admin

State Diagram

State Diagram-1: Psychiatrist

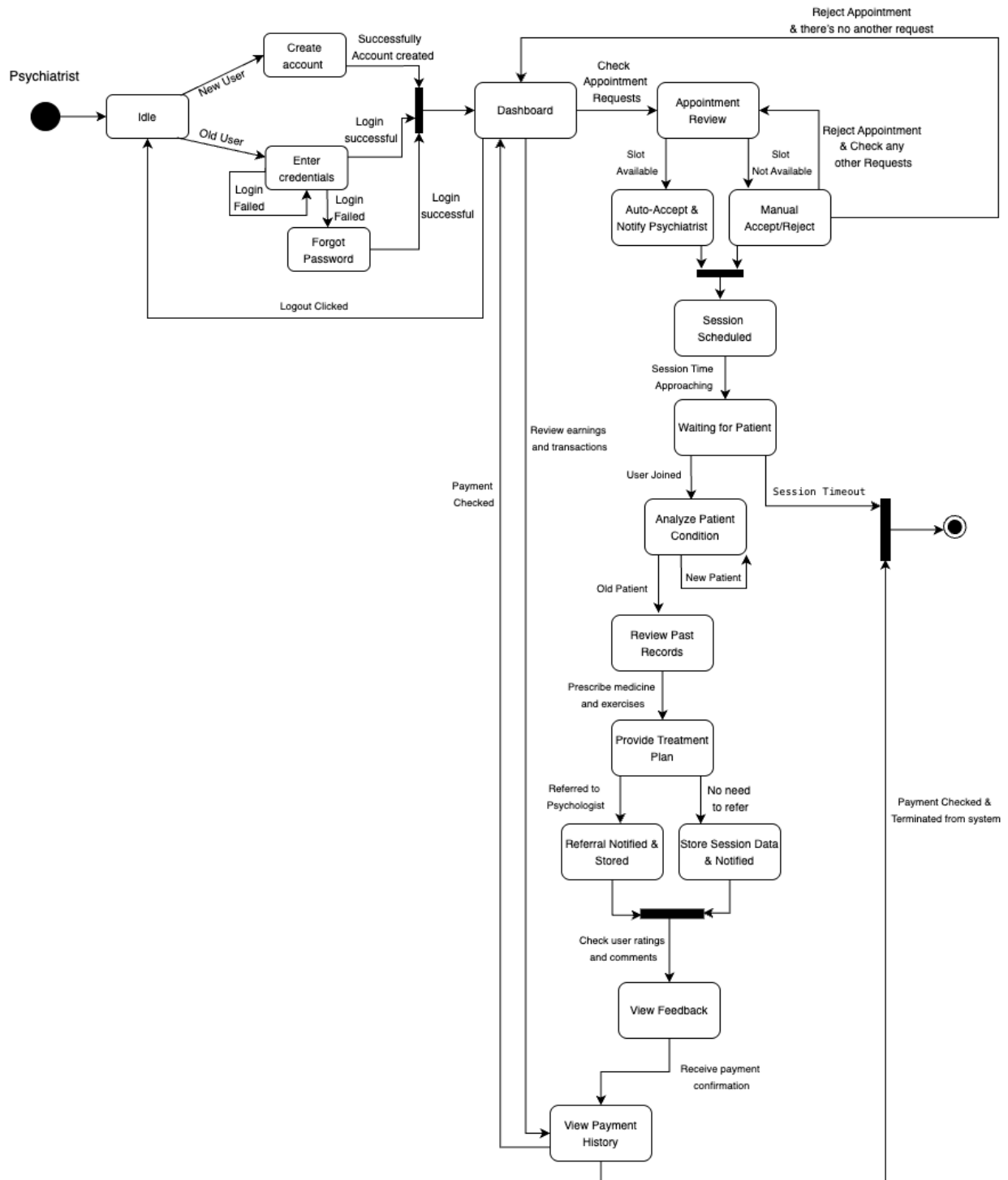


Figure-9: State Diagram for Psychiatrist

State Diagram-2: Psychologist

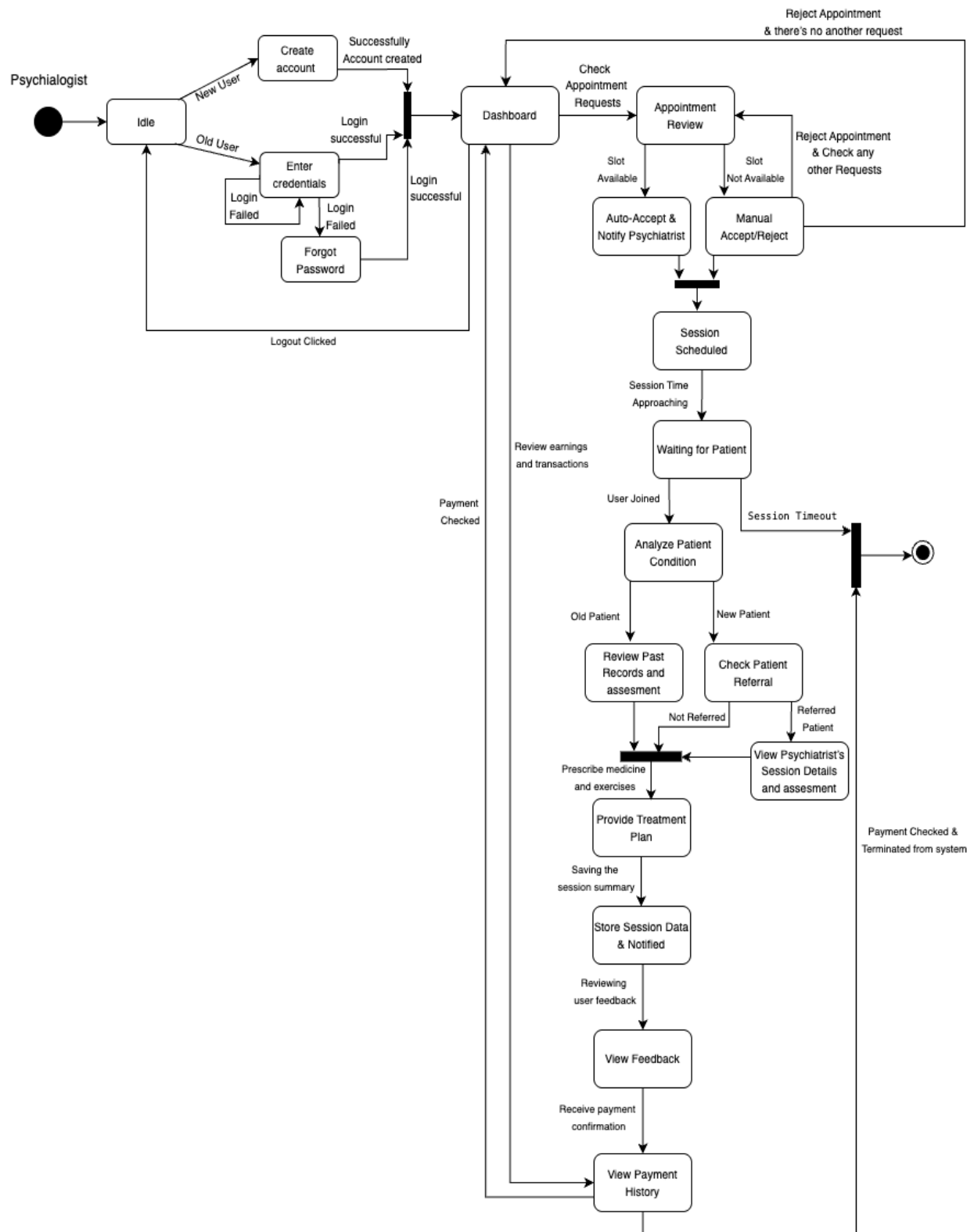


Figure-10: State Diagram for Psychologist

Figure-11: State Diagram for Patient



State Diagram-4: Admin

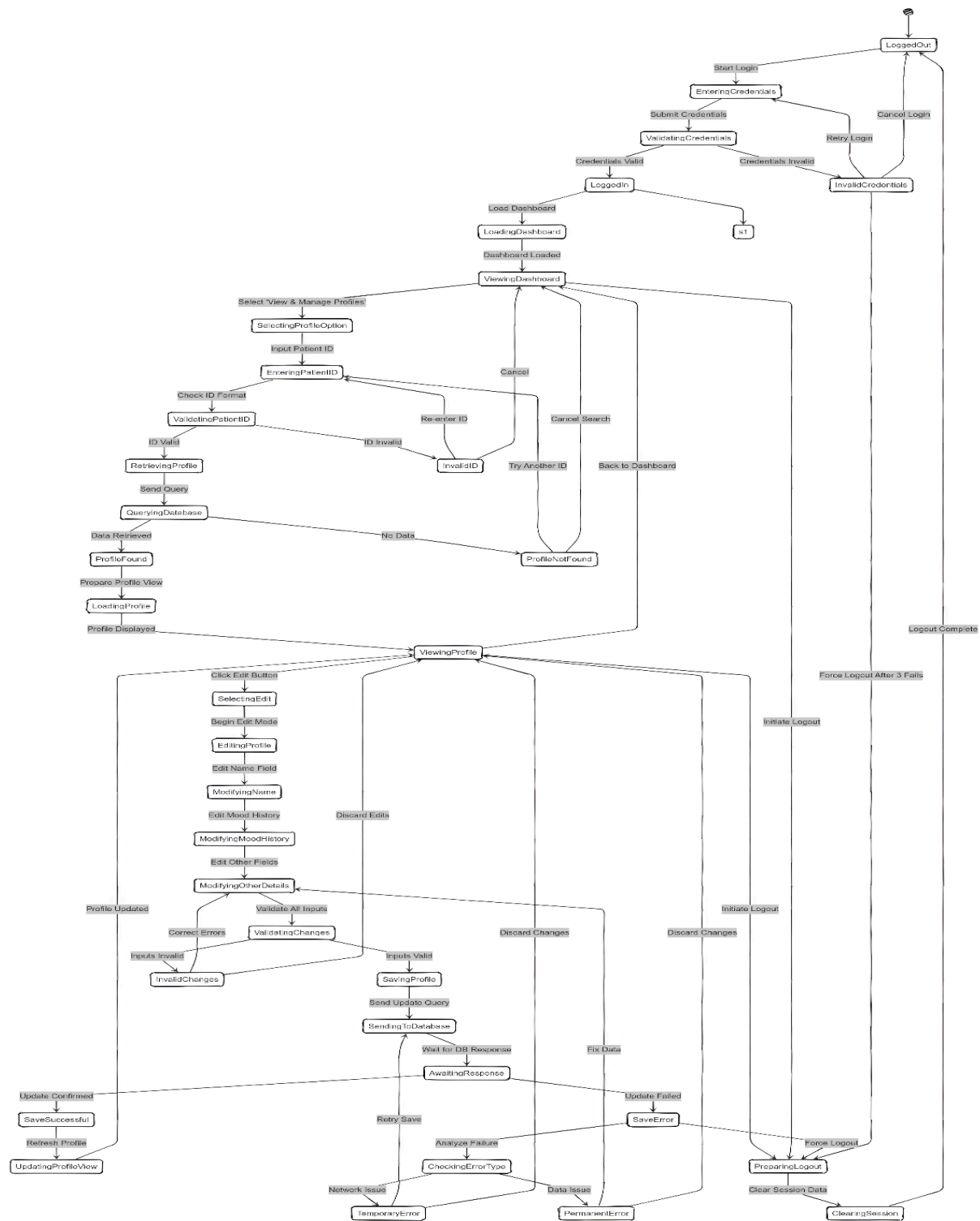


Figure-12: State Diagram for Admin

Sequence Diagram

Sequence Diagram-1: Psychiatrist

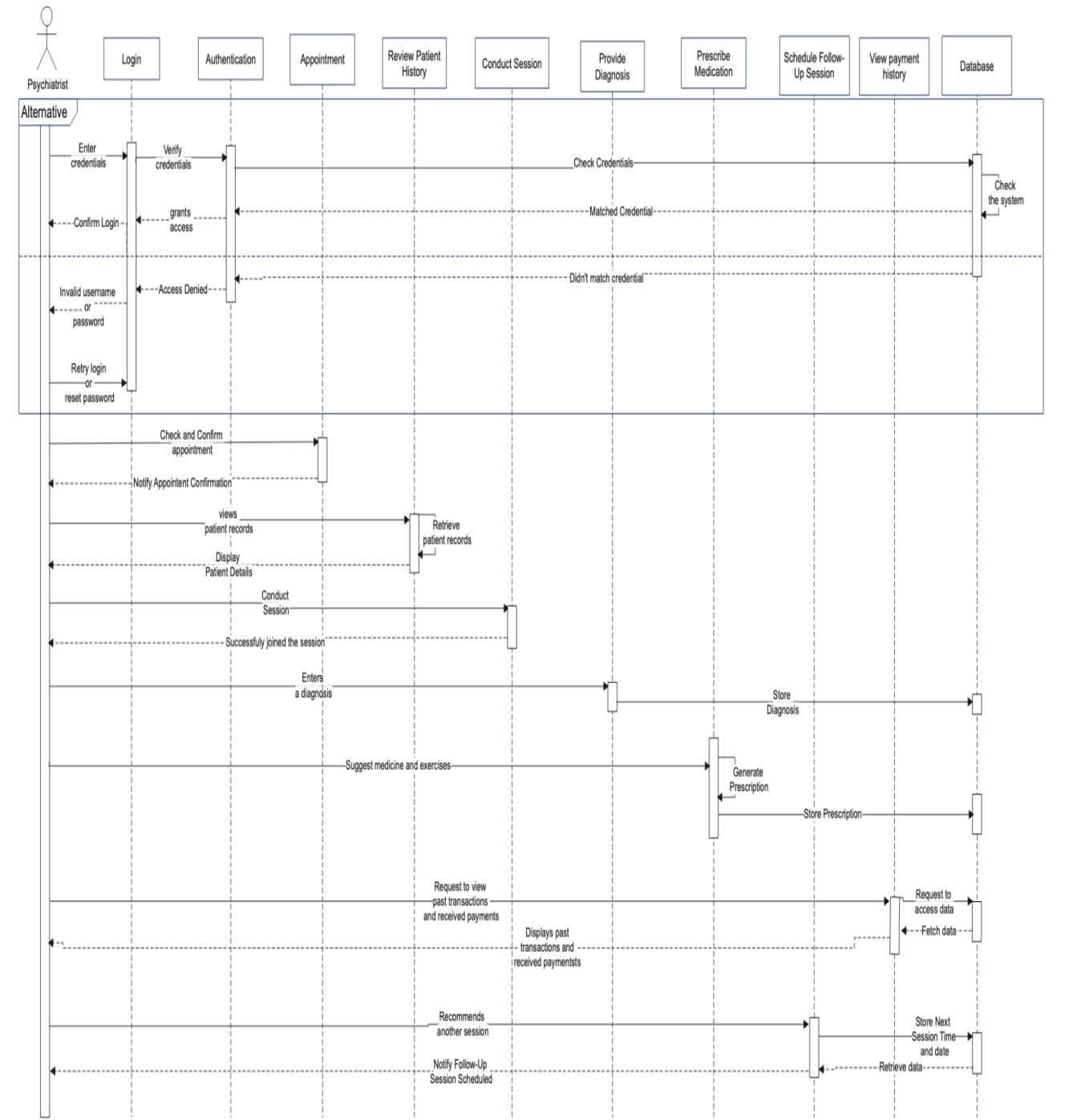


Figure-13: Sequence Diagram for Psychiatrist

Sequence Diagram-2: Psychologist

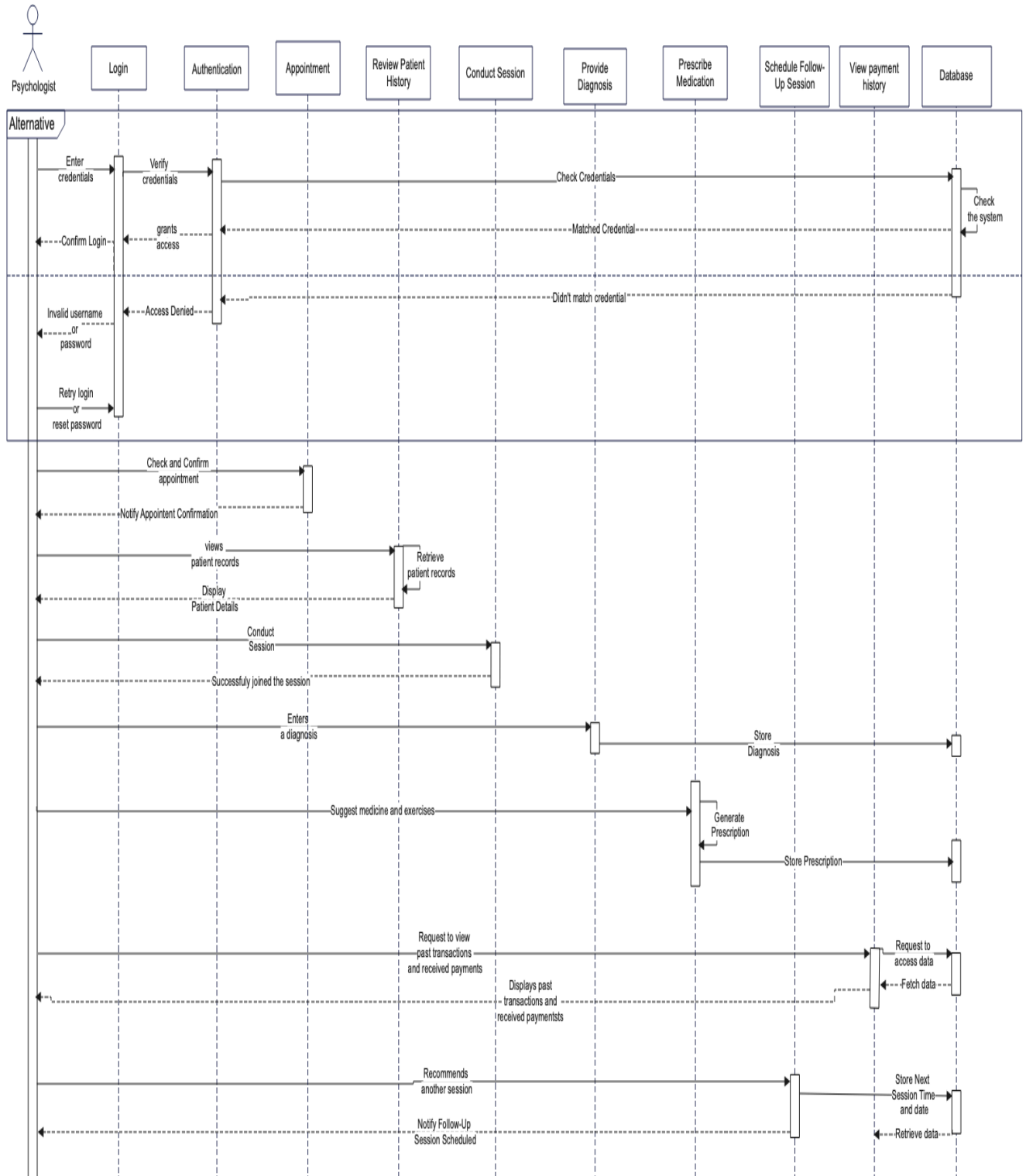


Figure-14: Sequence Diagram for Psychologist

Sequence Diagram-3: Patient

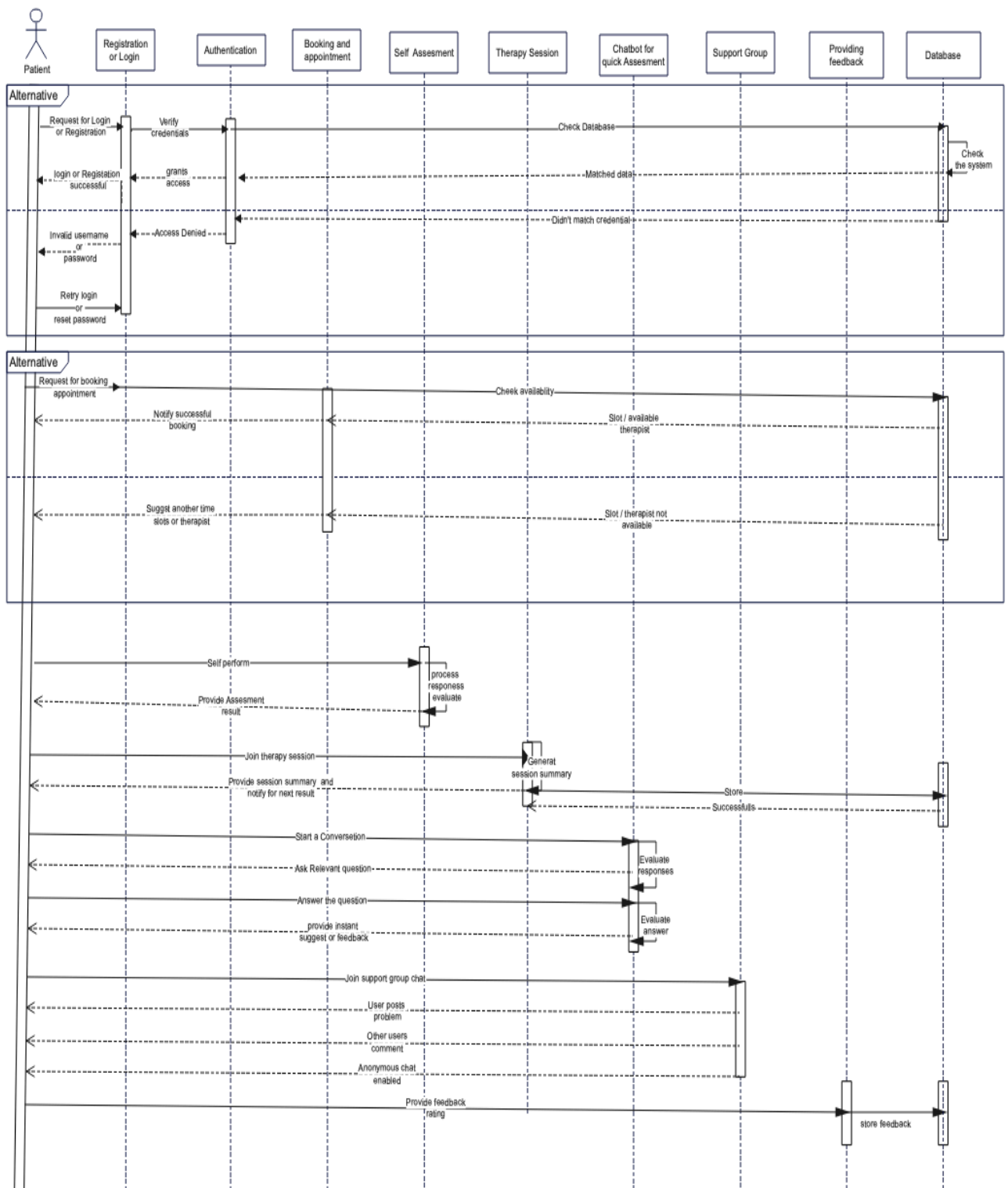


Figure-15: Sequence Diagram for Psychologist

Sequence Diagram-4: Admin

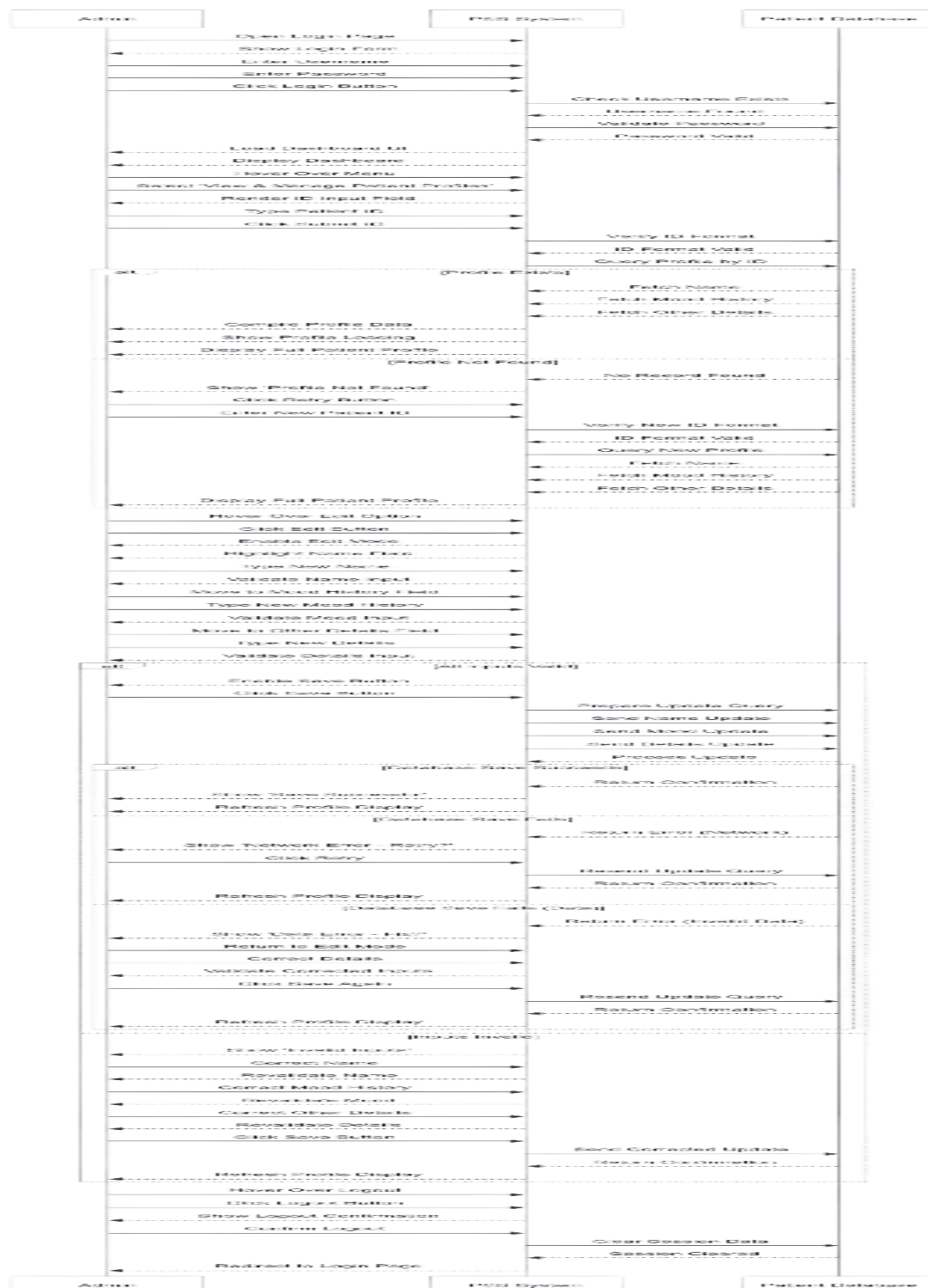


Figure-16: Sequence Diagram for Admin

Class Diagram

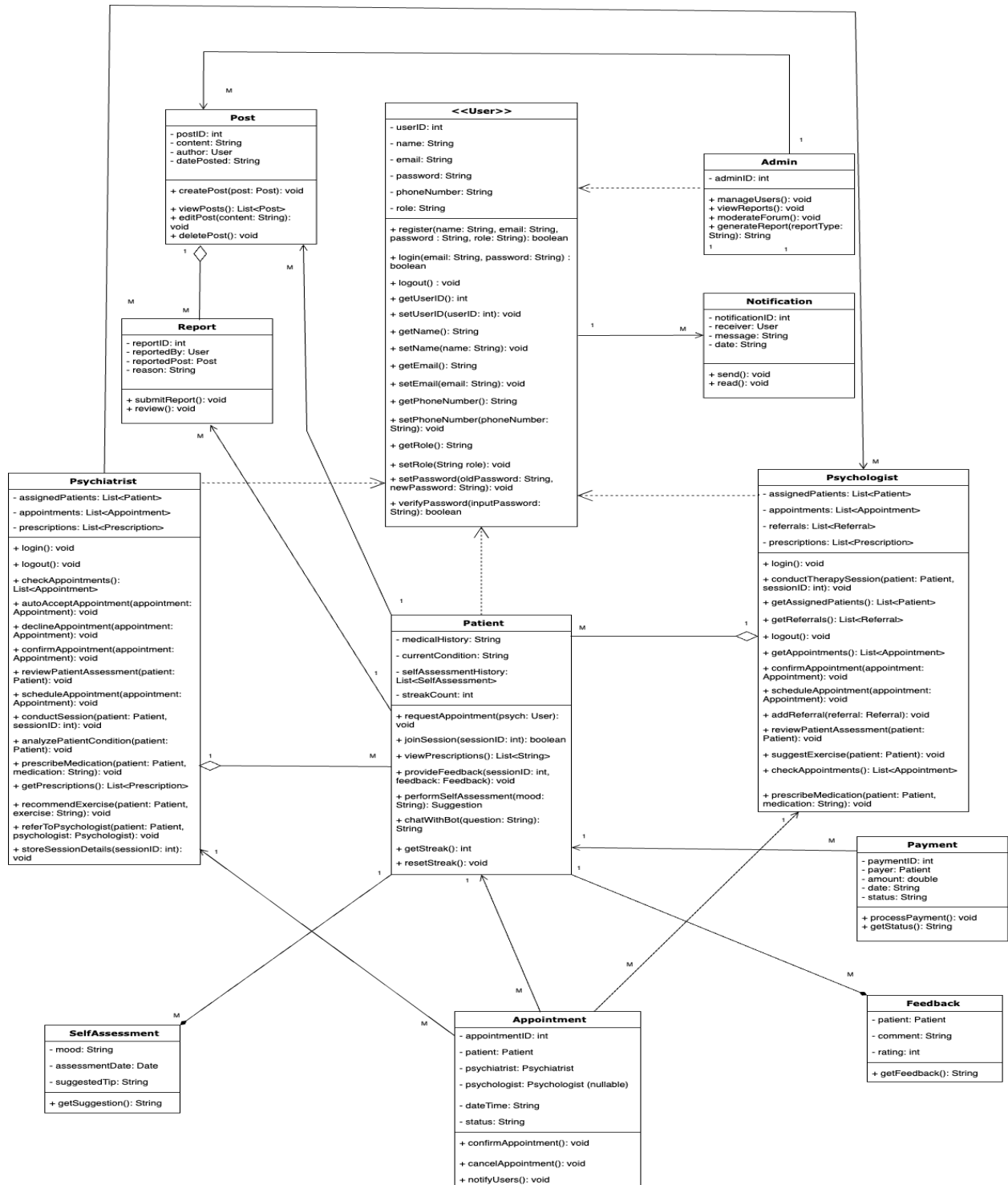


Figure-17: Class Diagram for Psychological Support System

Er Diagram

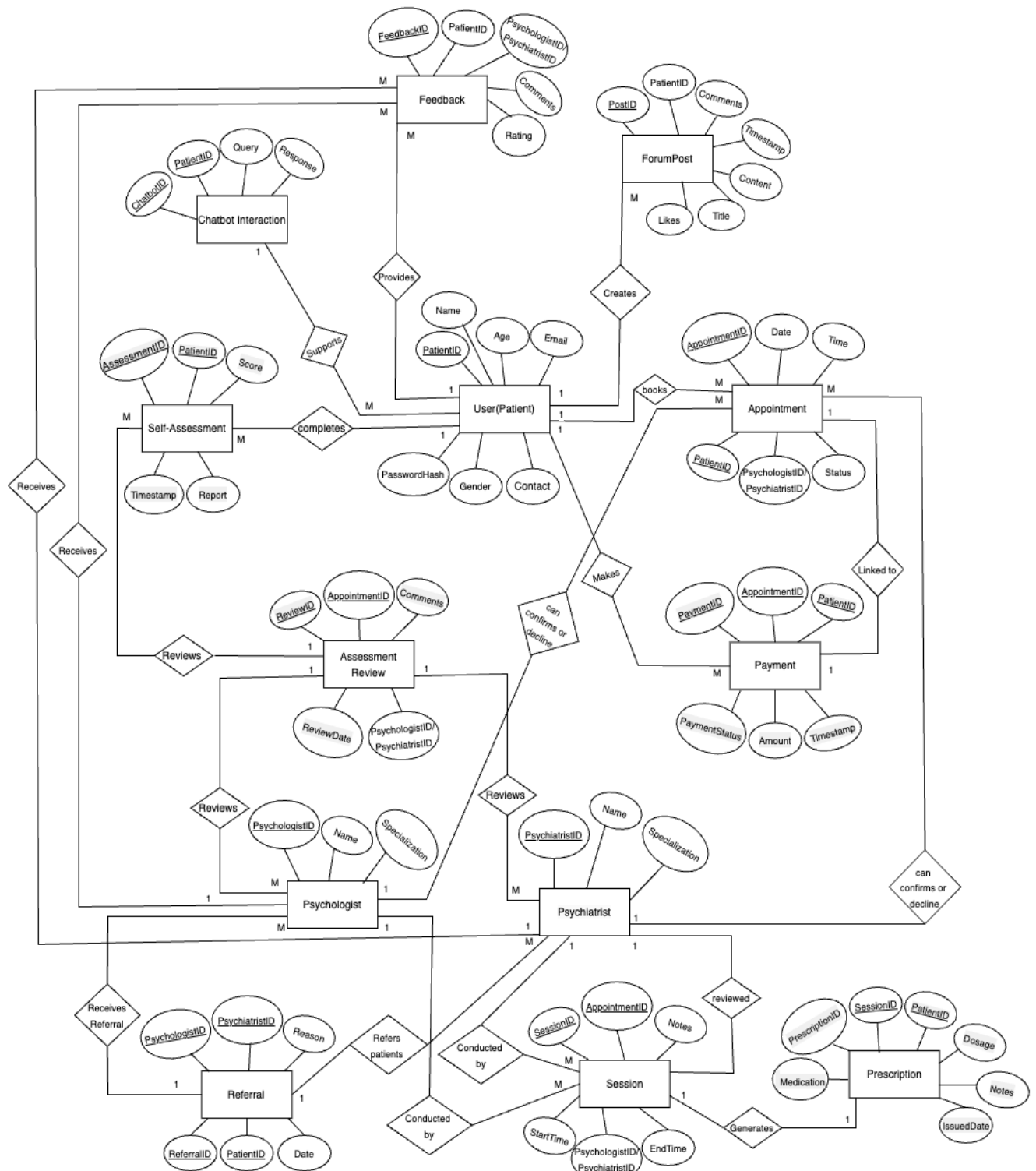


Figure-18: ER Diagram for Psychological Support System